

Operating Systems and Concurrency

Processes 1

COMP2007

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Recap

Last Lecture

- **Interrupts:**

- Change the normal flow of computation
- Switch from user to kernel mode

- **System calls:**

- Allow us to run kernel code to access services of the operating system
- Can be implemented in terms of interrupts

Goals for Today

Overview

- Introduction to **processes** and their **implementation**
- Process **states** and state **transitions**
- **System calls** for process management
- **The fork and exec pattern**

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- All the process control blocks are recorded in the **process table** - typically an array of PCBs.
- Every process has a unique **process identifier (PID)** used to associate it with its **PCB** - typically this is the corresponding array index.

Processes

Memory Image of Processes

- A **process' memory image** contains:
 - The program code (could be shared between multiple processes running the same code)
 - A data segment, stack and heap
- Every process has its own **logical address space**, in which the **stack** and **heap** are placed at **opposite sides** to allow them to grow

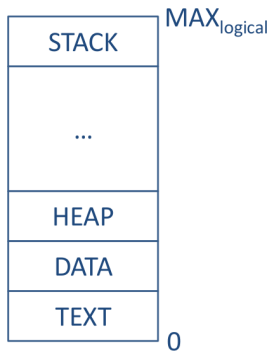
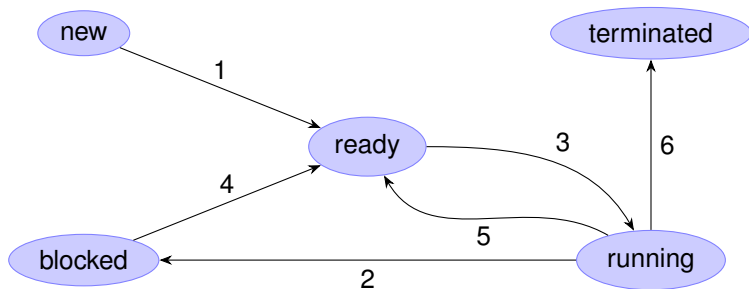


Figure: Representation of a process in memory

Process States and Transitions



Process States and Transitions

States

- A **new** process has just been created. It has a PCB and is waiting to be admitted, although it may not yet be in memory.
- A **ready** process is waiting for CPU to become available.
- A **running** process is currently having its instructions executed by the CPU.
- A **blocked** process cannot continue, e.g. is waiting for I/O
- A **terminated** process is no longer executable. The data structures - PCB - may be temporarily preserved.

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- A **suspended** process is swapped out (not discussed further)

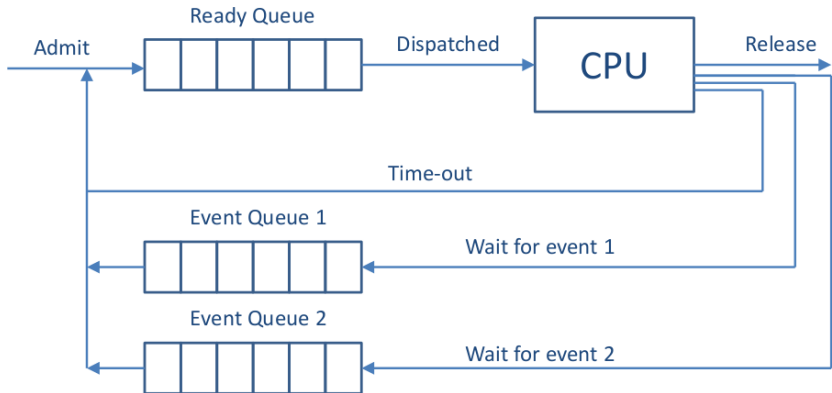
Process States and Transitions

Transitions

- State transitions include: important!!!
 - ① **New** → **ready**: admit the process and commit to execution
 - ② **Running** → **blocked**: e.g. process is waiting for input or carried out a system call
 - ③ **Ready** → **running**: the process is selected by the process scheduler
 - ④ **Blocked** → **ready**: event happens, e.g. I/O operation has finished
 - ⑤ **Running** → **ready**: the process surrenders the CPU, for example due to an **interrupt** or by **pause** give up
 - ⑥ **Running** → **terminated**: process has finished, e.g. program ended or exception encountered
- Interrupts and system calls drive these transitions.

Process States and Transitions

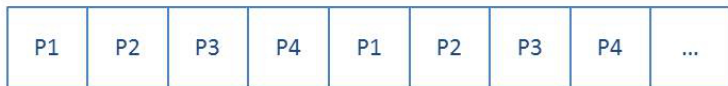
OS Queues



Context Switching

Multi-programming

- Modern computers are multi-programming systems
- Assuming a single processor system, the instructions of individual processes are executed **sequentially**
 - Multi-programming is achieved by interleaving the execution of processes, dividing the CPU time into time-slices
 - Control is exchanged between processes via a procedure known as **context switching**
 - A **trade-off** exists between the length of the **time-slice** and the **context switch time**
 - True parallelism requires hardware support



TIME



Context Switching

Multi-programming (Cont'd)

- When a **context switch** takes place, the system **saves the state** of the old process and **loads the state** of the new process (creates **overhead**)
 - **Saved** $\Rightarrow$ the process control block is **updated**
 - **(Re-)started** $\Rightarrow$ the process control block is **read**

Context Switching

Multi-programming (Cont'd)

Terminology

- The **response time** of a process is the ^{used} elapsed time from its creation before a process first gets access to the CPU
- The **effective utilisation** is the fraction of time the CPU is doing useful work - i.e. not context switching

Context Switching

Multi-programming (Cont'd)

Short time slices result in **good response times** but **low effective utilisation**. For example, assume both context switches and time slices take $1ms$. Then:

- It will take $99 \times (1 + 1) = 198ms$ for the last of 100 processes to start running.
- $\frac{1}{1+1} = 0.5$ of the CPU time is doing useful work.



TIME

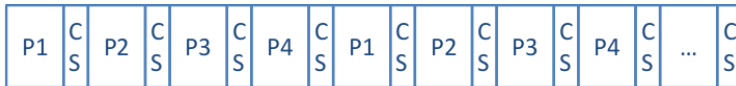


Context Switching

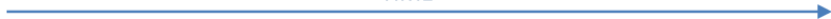
Multi-programming (Cont'd)

Long time slices result in **poor response times** but **better effective utilisation**. For example, assume context switches take $1ms$ and time slices are $100ms$. Then:

- It will take $99 \times (100 + 1) = 9999ms$ for the last of 100 processes to start running.
- $\frac{100}{1+100} = 0.99$ of the CPU time is doing useful work



TIME



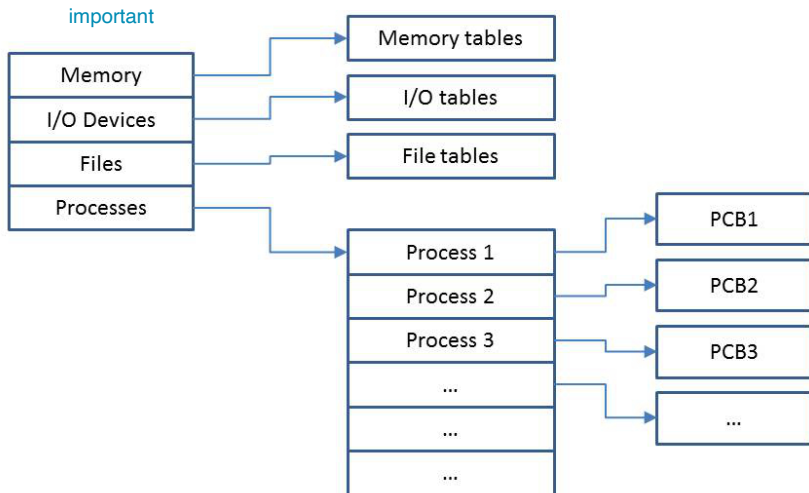
Context Switching

Multi-programming (Cont'ed)

- A **process control block** contains three types of **attributes**:
 - Process identification (PID, UID, Parent PID)
 - Process control information (process state, scheduling information, etc.)
 - Process state information (user registers, program counter, stack pointer, program status word, memory management information, files, etc.)
- **Process control blocks** are kernel data structures, i.e. they are **protected** and only accessible in **kernel mode**!
 - Allowing user applications to access them directly could compromise their integrity
 - The operating system manages them on the user's behalf through system calls (e.g. to set **process priority**)

Process Implementation

Tables and Control Blocks



Process Implementation

Tables and Control Blocks

- An operating system **maintains information** about the status of “resources” in **tables**
 - **Process tables** (process control blocks)
 - **Memory tables** (memory allocation, memory protection, virtual memory)
 - **I/O tables** (availability, status, transfer information)
 - **File tables** (location, status)
- The **process table** holds a **process control block** for each process, allocated upon **process creation**
- Tables are maintained by the **kernel** and are usually **cross referenced**

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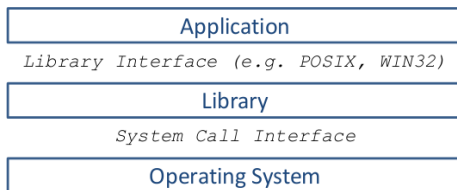
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- 6 Restore state from PCB (PC, registers, MMU, ...).
- 7 Return control to running the new process.

System Calls

Process Creation

- The true system calls are “**wrapped**” in the **OS libraries** following a well defined interface
- For example, on Unix-like operating systems `fork` is called to create a copy of a process. On Linux, the underlying system call used to implement `fork` is `clone`.



System Calls

Process Termination

- System calls via `exit` and `abort` can be used to explicitly notify the OS that the **process has terminated**
 - Resources must be de-allocated
 - Output must be flushed
 - Process admin may have to be carried out
- A system calls to terminate other processes:
 - UNIX/Linux: `kill()`
 - Windows: `TerminateProcess()`

System Calls

Waiting for Processes

- It is possible to **wait for a process and recover information about how it terminated**, using system calls such as `waitpid` to wait for a chosen PID.
- The operating system **must retain information about terminated processes to service wait calls**.
- Once a wait call has been serviced, the remaining control data can be cleaned up.

Process Creation in Linux

Fork

- fork() creates an **exact copy** of the current process
 - The first instruction carried out by the child is the first one after the `fork` call
- `fork()` returns the **process identifier (PID)** of the child process to the **parent process**.
- `fork()` **returns 0** to the **child process**.

Process Creation in Linux

The Fork and Exec Pattern

A common pattern is the following sequence:

- 1 Call `fork()` to create an exact copy of the current process.
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This is a typical pattern of calls in a Unix shell such as `bash`.

Processes

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#include <unistd.h>
#include <sys/wait.h>

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Test your understanding

- Why might you run `fork` without running a subsequent `exec`?
- Do you always need to call `exit` to end a process?
- Why does a process control block contain data about register contents?
- Why might it be useful to retain a process control block for a terminated process?

1. The child process can continue executing the same program as the parent without loading a new program.
2. No. A process also terminates when `main()` returns or reaches its end.
3. The OS must save register contents during a context switch so the process can later resume correctly.
4. To preserve termination information, such as the exit status, until the parent collects it with `wait()` or `waitpid()`.